

Vector-Valued Functions



Position, velocity, and acceleration vectors

- 1 A particle's position is given by $\vec{r}(t) = \langle t^2, t^3 \rangle$. Find the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.
 - 2 For the same particle, find the speed at $t = 2$.
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Integrating vector-valued functions

- 3 Given $\vec{v}(t) = \langle 3t^2, 2t \rangle$ and initial position $\vec{r}(0) = \langle 1, 4 \rangle$, find $\vec{r}(t)$.
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Total distance traveled

- 4 A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, \pi]$. Find the total distance traveled.
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Vectors and parametric motion combined

- 5 A particle moves along $x(t) = e^t$, $y(t) = e^{-t}$. Find the velocity vector and speed at $t = 0$.
 - 6 For the particle above, find the magnitude of the acceleration vector at $t = 0$.
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Unit tangent vectors

- 7 For $\vec{r}(t) = \langle t, t^2 \rangle$, find the unit tangent vector $\vec{T}(t)$ at $t = 1$.
 - 8 Explain why $\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$ always has magnitude 1.
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Motion on a circle and centripetal acceleration

- 9 A particle moves along $\vec{r}(t) = \langle 3\cos t, 3\sin t \rangle$. Find $\vec{v}(t)$, $\vec{a}(t)$, and verify that $\vec{a}(t)$ points toward the origin (is a scalar multiple of $-\vec{r}(t)$).
 - 10 For the same particle, verify that $\vec{v}(t) \cdot \vec{r}(t) = 0$ for all t , and interpret this geometrically.
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Displacement vs. distance for vector motion

- 11 A particle has position $\vec{r}(t) = \langle t^2, t \rangle$ for $t \in [0, 2]$. Find the displacement vector from $t = 0$ to $t = 2$.
- 12 Set up (but do not evaluate) the integral for the total distance traveled by the particle $\vec{r}(t) = \langle t^2, t \rangle$ over $t \in [0, 2]$.
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Vector operations review

- 13 Given $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, find $2\vec{u} - 3\vec{w}$.
- 14 Find the dot product $\vec{u} \cdot \vec{w}$ for $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, and state whether the vectors are perpendicular.
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Curvature-related speed problems

- 15 A particle moves along $\vec{r}(t) = \langle t^3, t^2 \rangle$. Find the times t when the particle's speed is exactly $\sqrt{13}$.
- 16 For the particle $\vec{r}(t) = \langle t, \ln t \rangle$ ($t > 0$), find the velocity vector and speed at $t = 1$.
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Projectile motion as a vector-valued function

- 17 A projectile is launched with position $\vec{r}(t) = \langle 20t, 30t - 4.9t^2 \rangle$ (meters, seconds). Find the velocity vector $\vec{v}(t)$ and the time when the projectile reaches maximum height (where the vertical velocity component is 0).
- 18 For the same projectile, find the speed at $t = 0$ (launch) and interpret its two components.
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Finding when a particle is at rest

- 19 A particle moves along $\vec{r}(t) = \langle t^3 - 3t, t^2 - 4 \rangle$. Find all times t when the particle is momentarily at rest (both velocity components equal 0 simultaneously).
- 20 For the same particle, find the times when only the horizontal velocity component is 0.
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Vector-valued functions and arc length

- 21** State the arc length formula for a vector-valued function $\vec{r}(t) = \langle x(t), y(t) \rangle$ on $[a, b]$, and explain how it relates to the speed $|\vec{v}(t)|$.
- 22** Find the arc length of $\vec{r}(t) = \langle 4t, 3t \rangle$ for $t \in [0, 5]$ using the formula above.